CAPSTONE PROJECT

CHAPTER 3 - REQUIREMENTS & METHODOLOGY

Student / Group Template

Project Title	TrackWIL
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3.1 Introduction

Introduce the approach used to gather requirements and develop the system.

This chapter explains how the requirements for TrackWIL will be identified and how the system will be developed. TrackWIL is a proposed web-based system designed to improve the management of Work-Integrated Learning (WIL) documents and student progress.

Currently, students may need to complete and sign daily WIL documents, send them to workplace mentors for verification and wait for the signed documents to be returned. This manual process can lead to delays, lost documents and difficulties tracking student progress.

TrackWIL aims to provide a central platform where students can submit daily work records, mentors can review and verify submissions, and coordinators and lecturers can monitor student progress. This chapter describes the stakeholders, system requirements, requirements gathering process, development methodology, user stories, system architecture and ethical considerations that will guide the development of the system.

3.2 Stakeholders and Users

Identify primary and secondary stakeholders and their needs.

Stakeholders are people or organisations affected by TrackWIL, while users are the people who interact directly with the system.

Primary stakeholders and users

Students: Students will use TrackWIL to create and submit daily WIL records, upload supporting documents and check whether their submissions have been approved or require corrections. Their main need is a simple and reliable way to manage their WIL documentation without relying entirely on paper documents and email.

Workplace Mentors: Mentors will review records submitted by students assigned to them. They will be able to verify submissions, approve correct records and request corrections when necessary. Their main need is an organised way to review and confirm student work.

Coordinators: Coordinators will monitor student submissions and identify outstanding records. They need an overview of student documentation so that they can follow up on missing submissions and help ensure that WIL requirements are being completed.

Lecturers: Lecturers will monitor student progress and identify students who may be falling behind. They need access to relevant progress information to provide academic support and follow up when necessary.

Secondary stakeholders

Training Institution: The institution requires students to complete their WIL activities and maintain the necessary documentation. TrackWIL can support the institution by making records easier to manage and monitor.

Development Team: The development team is responsible for analysing requirements, designing, developing and testing the system. The team needs clear requirements, suitable technologies and effective collaboration to deliver the project successfully.

Project Supervisor: The project supervisor oversees the academic project and checks whether the system meets the project objectives and assessment requirements.

3.3 Functional Requirements

List what the system must do. Give each requirement an ID, description and priority.

ID	Description	Priority
FR-01	The system must allow authorized users to log in using valid credentials.	High
FR-02	The system must provide different access permissions for students, mentors, coordinators and lecturers.	High
FR-03	Students must be able to create daily WIL records describing the work completed.	High
FR-04	Students must be able to submit completed daily records for review.	High
FR-05	Students must be able to upload supporting documents or evidence related to their daily records.	High
FR-06	Workplace mentors must be able to view records submitted by their assigned students.	High
FR-07	Workplace mentors must be able to approve records or request corrections with feedback.	High
FR-08	The system must display the status of each record as pending, approved or requiring corrections.	High
FR-09	Students must be able to view their previous submissions and their status.	Medium
FR-10	Coordinators must be able to identify students with outstanding or missing records.	High
FR-11	Lecturers must be able to monitor student progress and identify students who are falling behind.	High
FR-12	The system must record the dates and times of submissions and verifications.	High
FR-13	Authorized users must be able to view relevant student activity and progress information.	Medium

FR-14	Users must be able to log out of their accounts securely.	High

3.4 Non-Functional Requirements

Document usability, performance, security, reliability, maintainability and other quality requirements.

- **Usability:** TrackWIL should have a simple, understandable interface that allows users to navigate their dashboards, submit records and check document statuses easily.
- **Performance:** The system should respond to normal user actions within an acceptable time. Dashboard information and document submission functions should load without unnecessary delays. The team should define and test specific response-time targets.
- **Security:** The system must protect user accounts and information through appropriate authentication, role-based access control and secure handling of uploaded documents.
- **Reliability:** Submitted records and supporting documents should be stored reliably. Backup and recovery measures should be considered to reduce the risk of losing important information.
- **Maintainability:** The application should use an organised project structure and clear documentation so that developers can correct errors, update features and maintain the system.
- **Compatibility:** TrackWIL should work on commonly used modern web browsers and adapt to different screen sizes, including computers, tablets and mobile phones.
- **Data Integrity:** The system should maintain accurate and consistent records. Submissions must be linked to the correct students and placements, and only authorised users should be able to make permitted changes.
- **Privacy:** Personal information and WIL documents must only be accessible to authorised users who need the information to perform their responsibilities.

3.5 Requirements Gathering

Explain how requirements were identified, such as interviews, observation, questionnaires, document analysis or scenario analysis.

Requirements gathering is the process of identifying and understanding what a system must do and what users need from it. For the TrackWIL project, requirements will be identified by analysing the project scenario, reviewing the assignment brief, identifying the system users, and discussing the required features with the project team.

Scenario Analysis

The TrackWIL scenario will be analysed to understand the problems with the current manual WIL documentation process. Students currently complete daily work documents and send them to workplace mentors for signing and verification. This process can cause delays, and documents may be lost or accidentally

deleted. TrackWIL aims to reduce these problems by allowing students to submit their daily work records digitally and making it easier for authorised users to monitor and verify submissions.

Assignment Brief and Document Analysis

The project brief and assignment requirements will be reviewed to identify the main features and expectations of TrackWIL. These requirements will guide the design and development of the system, including user login, daily work record submissions, document uploads, mentor verification, and progress monitoring.

Stakeholder Analysis

The needs of the main TrackWIL users will be identified to ensure that the system supports their responsibilities.

- **Students:** Need to record daily work activities, upload supporting documents, and check the status of their submissions.
- **Workplace mentors:** Need to view students' submissions, verify completed work, and provide feedback when corrections are required.
- **Coordinators:** Need to monitor submissions and identify outstanding or missing records.
- **Lecturers:** Need to check student progress and identify students who are falling behind with their WIL requirements.

Project Team Discussions

The project team will discuss the identified requirements and agree on the features that should be included in TrackWIL. These discussions will help the team organise the requirements according to their importance and determine which features should be developed first.

Requirements Review

The identified requirements will be reviewed to ensure that they are clear, realistic, and relevant to the TrackWIL problem. The requirements will then be documented in the Software Requirements Specification (SRS) and used to guide the system design, development, and testing.

The requirements gathering process will help ensure that TrackWIL addresses the problems associated with manual WIL documentation and meets the needs of its intended users.

3.6 Development Methodology

Describe the chosen SDLC/Agile approach and justify why it fits this project.

The proposed methodology for TrackWIL is Agile. Agile is an iterative approach that allows software to be developed in smaller stages, tested regularly and improved as the project progresses.

- **Planning:** The team identifies the problem, defines the scope and agrees on the project objectives.
- **Requirements Analysis:** The team identifies the users and documents the functional and non-functional requirements.
- **System Design:** The team prepares the system architecture, UML diagrams, database design and user interface wireframes.
- **Development:** The team builds the system in smaller parts, starting with essential features such as login, student dashboards and daily record submission.
- **Testing:** Each developed feature is tested to identify errors and confirm that it meets its requirements.
- **Review and Improvement:** The team reviews completed work, discusses problems and improves the system based on testing results and available feedback.
- **Implementation and Maintenance:** The application is prepared for use, and problems or improvements are addressed where required by the project scope.

Justification for Agile

Agile is suitable for TrackWIL because the system contains several connected features that can be developed and tested separately. For example, the team can develop student record submission before completing the mentor verification and progress monitoring functions.

Agile allows problems to be identified early and makes it easier to improve features as the team learns more about the system. It also supports teamwork because members can work on different tasks and regularly review their progress.

3.7 Use Cases / User Stories

Document key interactions between users and the system.

1. **US-01: Student Login**

As a student, I want to log in securely so that I can access my WIL records and dashboard.

2. **US-02: Daily Record Submission**

As a student, I want to create and submit a daily WIL record so that my workplace activities are documented and stored in one place.

3. **US-03: Upload Supporting Documents**

As a student, I want to upload supporting evidence so that my workplace activities can be reviewed without relying entirely on paper documents or email attachments.

4. **US-05: Mentor Verification**

As a workplace mentor, I want to review student submissions and approve them or request corrections so that the records are properly checked.

5. **US-06: Monitor Outstanding Records**

As a coordinator, I want to identify students with missing or outstanding records so that I can follow up and help ensure that the required documentation is completed.

6. **US-07: Monitor Student Progress**

As a lecturer, I want to view relevant student progress information so that I can identify students who are falling behind and provide support.

3.8 System Architecture

Describe the proposed architecture and major components.

TrackWIL is proposed as a web-based application that connects users to a central system for managing WIL records and documents. The system architecture will be organised into three main layers: the presentation layer, application layer and data layer.

Presentation Layer

The presentation layer is the part of the system that users interact with through a web browser. It will provide the login page and dashboards for students, workplace mentors, coordinators and lecturers.

Students will use this layer to create daily records, upload evidence and check submission statuses. Mentors will use it to review and verify submissions, while coordinators and lecturers will use their dashboards to monitor student progress.

Application Layer

The application layer handles the system's main operations and business rules. These include authentication, checking user permissions, processing daily record submissions, managing document verification and updating submission statuses.

This layer must ensure that users can only perform actions permitted by their roles. For example, a student should not be able to approve their own submission as a mentor.

Data Layer

The data layer stores and manages the information required by TrackWIL. This may include user accounts, WIL placements, daily records, document references and verification history.

A relational database such as PostgreSQL could be used to maintain relationships between these records. Uploaded documents would be stored using an appropriate file storage service, with access restricted to authorised users.

Communication Between the Layers

When a student submits a daily record, the presentation layer sends the information to the application layer. The application layer checks the request, verifies the user's permissions and saves the record through the data layer. The system then returns a response so that the student can see whether the submission was successful.

When a mentor reviews the record, the same general process is used to retrieve the relevant information and save the verification decision.

This architecture separates the user interface, application operations and data storage, making the system easier to organise, test and maintain. The final architecture diagram will show these layers and the connections between them.

The exact frameworks, database and storage services will be confirmed according to the technology stack selected by the development team.

3.9 Ethical, Privacy and Security Considerations

Explain responsible handling of users, data, consent, privacy and security.

User Authentication and Access Control

TrackWIL must use appropriate authentication and role-based permissions to protect user accounts. Students should access their own records, mentors should access records assigned to them, and coordinators and lecturers should only access information required for their responsibilities.

Data Accuracy and Accountability

The system should maintain accurate submission and verification information. Important actions, such as submitting a record, approving it or requesting corrections, should be recorded where appropriate. This will help users understand the history of a document and reduce confusion about whether it has been reviewed. Verification must also be performed by the authorised mentor rather than being treated as automatically approved.

Consent and Responsible Data Use

Users should receive appropriate information about what personal information the system collects and why it is needed. Where consent is the appropriate lawful basis for processing, it must be obtained in accordance with applicable requirements. Personal information should only be used for authorised purposes.

Privacy and Protection of Personal Information

The system should collect only the information necessary for WIL administration and monitoring. Personal information must be handled responsibly and in accordance with applicable institutional policies and South African data protection requirements, including the Protection of Personal Information Act (POPIA), where applicable.

3.10 Chapter Summary

Summarise how the requirements and methodology guide development.

This chapter presented the requirements and proposed development methodology for TrackWIL. It identified the main stakeholders and users, including students, workplace mentors, coordinators and lecturers, and explained their needs.

The functional requirements described the features needed for login, daily record submission, document uploads, mentor verification and progress monitoring. The non-functional requirements addressed usability, performance, security, reliability, compatibility, maintainability, data integrity and privacy.

The chapter also explained how requirements can be gathered and reviewed, justified the proposed use of Agile over Waterfall, and documented user stories and the main daily record verification use case. Finally, it described the proposed three-tier system architecture and discussed ethical, privacy and security considerations.

These requirements and design decisions will guide the detailed Software Requirements Specification, the system design in Chapter 4, and the implementation and testing of TrackWIL.